Coal Transportation by Indian Railways

WAGONS, RAKES, CAPACITIES & PENALTY MECHANISM

Types of Wagons Used for Coal Transportation

Wagon Type	Full Form	Typical Use	Loading Method
BOXN	Bogie Open eXtra capacity	Thermal coal, bulk coal	Top loading
BOXNHL	BOXN High Load	High-density coal	Top loading
BOBRN	Bogie Open Bottom Rapid	Power plants (fast unloading)	Bottom discharge
BOBYN	Bogie Open Bottom (New)	High-capacity coal rakes	Bottom discharge

Wagon Type	Carrying Capacity (MT)	Gross Tare Weight
BOXN	63 – 65 MT	~23 MT
BOXNHL	70 – 72 MT	~25 MT
BOBRN	68 – 70 MT	~24 MT
BOBYN	72 – 75 MT	~25 MT

BOXN Wagon
(Most common coal wagon)

BOXN – Bogie Open High-Capacity Wagon

- Open-top wagon
- Top loading & top unloading
- Widely used for coal, iron ore, minerals

Typical capacity: 63–65 MT
Used by: Mines, sidings, industries

BOXNHL Wagon
(High-load version of BOXN)

BOXNHL – Bogie Open Extra High Load

- Improved design over BOXN
- Carries more coal per wagon
- Same loading method (top loading)

Typical capacity: 70–72 MT
Advantage: Fewer wagons → more payload

BOBRN Wagon
(Bottom discharge wagon; fast unloading)

BOBRN – Bogie Open Bottom Rapid

- Bottom doors open automatically
- Used at power plants with track hoppers
- Very fast unloading

Typical capacity: 68–70 MT
Used mainly for: Thermal power plants

BOBYN Wagon
(New-generation bottom discharge wagon)

BOBYN – Bogie Open Bottom (New Design)

- Advanced version of BOBRN
- Higher axle load
- Better safety & faster unloading

Typical capacity: 72–75 MT
Used for: High-volume coal corridors

BOBSN Wagon
(Special bottom discharge variant)

BOBSN – Bogie Open Bottom Special

- Bottom discharge
- Less common today
- Older design compared to BOBYN

Typical capacity: ~65–68 MT

Types of Rakes

Rake Type	No. of Wagons	Typical Use
Full Rake	59 wagons	Power Plants, Big Mines
Half Rake	29 wagons	Medium Industries
Mini Rake	15 wagons	Small Consumers

Types of Rakes and Penalties

Code	Full Form	Applied When
POLA	Punitive Over Loading (Average)	Rake average exceeds PCC
POL1	Punitive Over Loading (Individual)	Wagon exceeds PCC
PLO	Penal Loading Overcharge	Non-punitive excess
ULC	Uneven Loading Charge	Axle imbalance
SPL	Spillage / Loose Loading	Coal falling / unsafe heap
DEM	Demurrage	Time delay
WMC	Weighment Charge	Weighbridge use
MCF	Moisture Content Factor	Wet coal disputes

Penalty / Charges

POL1: Punitive Overloading (Individual Wagon)

Applied When:

- A wagon exceeds its permissible capacity

Example:

- Capacity: 70 MT
- Loaded: 73 MT
- Excess: 3 MT

Penalty Calculation:

Excess MT × Distance × Punitive Rate

Penalty / Charges

POLA: Punitive Overloading (Average)

Applied When:

- Average load of entire rake exceeds permissible capacity
- Even if individual wagons are within limit

Example:

- Total load: 4,090 MT
- Wagons: 58

Penalty:

Average = 70.52 MT → POLA applicable

Penalty / Charges

PLO: Proportionate Loading Overcharge

Nature:

- Not a penalty
- Extra freight charge

Applied When:

- Load exceeds base chargeable weight
- But within permissible capacity

Example:

- Base: 68 MT
- Loaded: 69 MT

Penalty:

Extra 1 MT charged at normal freight rate

Penalty / Charges

ULC: Uneven Loading Charge

Applied When:

- Coal is unevenly distributed
- Axle load imbalance detected

Consequences:

- Wagon detention
- Re-loading required
- Additional charges

Penalty / Charges

SPL: Spillage / Loose Loading

Applied When:

- Coal loaded above wagon rim
- Coal spilling on track

Charges Include:

- Cleaning charges
- Environmental penalty
- Detention charges

Penalty / Charges

DEM: Demurrage Charges

Meaning:

- Delay beyond free loading/unloading time

Typical Free Time:

- Loading: 3 hours
- Unloading: 1-2 hours

Charged:

- Per wagon per hour

Best Practices to Avoid Penalties

- Load 1–2 MT below PCC
- Ensure uniform distribution
- Avoid loading during heavy rain
- Trim heap below wagon rim
- Maintain wagon-wise loading records

Key Points

- ✓ Know your wagon capacity clearly
- ✓ Monitor both **individual** and **average** loading
- ✓ POL1 and POLA are the costliest penalties
- ✓ Most penalties are **preventable with discipline**